

Assignment 3

Deadline: 10/21/2025, h: 11:59pm

- This assignment sheet has 4 exercises. Submit your solution via Gradescope. See the syllabus for the late submission policy.
- You are allowed to discuss the assignment with others but the write-up must be individual work. Please mention in your write-up all the people you have discussed the solution with.

a) Zero-sum games

Problem 1. Consider the following class of zero-sum games, which can be naturally applied, e.g., to political campaigns, marketing, auction and procurement decisions, and public policy or humanitarian operations.

Two players simultaneously allocate limited resources across several “battlefields”. To each battlefield i , we associate a value v_i . On each battlefield i the player allocating the most wins v_i ; the player allocating the least loses v_i ; and if they both allocate the same amount of resources they both neither win nor lose.

Consider the following instance. There are two battlefields, with values $v_1 = 3$ and $v_2 = 4$. The first player has a total of 3 units of resources; the second player has a total of 4 units of resources. They can each allocate an integer number of resources to each of the two battlefields.

For instance, suppose that the first player allocates 1 unit of resource to the first battlefield and 2 units to the second battlefield; and the second player allocates 2 units of resources to each of the two battlefields. We denote these strategies by $(1, 2)$ and $(2, 2)$. Then, the second player wins 3 units (3 from the first battlefield, 0 from the second), and symmetrically the first player loses 3. If conversely the first player plays $(3, 0)$ with probability $1/2$ and $(0, 3)$ with probability $1/2$, while the second player always plays $(3, 1)$, then each of the following outcomes have $1/2$ probability of happening:

- Player 1 plays $(3, 0)$ and player 2 plays $(1, 3)$, and the outcome for player 2 is $-3 + 4 = 1$.
- Player 1 plays $(0, 3)$ and player 2 plays $(1, 3)$, and the outcome for player 2 is $3 + 0 = 3$.

So the expected outcome for player 2 is 2 (and symmetrically, for player 1 it is -2).

- Write the linear programs for computing the optimal strategies in the two-player game corresponding to the instance above. Compute their optimal solutions.
- Now consider a generic instance with k battlefields with integer values $v_1, \dots, v_k \geq 1$, and where each of the two player has an integer budget $b_1, b_2 \geq 1$ to split (in integer quantities) among the k battlefields. Write linear programs to compute the equilibria.

Problem 2: Recall the LPs that compute the optimal strategies (for the column and the row players, respectively) of the players in the Rock-Paper-Scissors game:

$$\begin{array}{ll}
 \max v & \min u \\
 \begin{array}{ll}
 [Ax]_R & \geq v \\
 [Ax]_S & \geq v \\
 [Ax]_P & \geq v \\
 x_P + x_R + x_S & = 1 \\
 x_P, x_R, x_S & \geq 0
 \end{array} & (1) \qquad \begin{array}{ll}
 [y^T A]_R & \leq u \\
 [y^T A]_S & \leq u \\
 [y^T A]_P & \leq u \\
 y_P + y_R + y_S & = 1 \\
 y_P, y_R, y_S & \geq 0
 \end{array} & (2)
 \end{array}$$

Show that the value of their optimal solutions coincide. (*Write the duals of the two problems and show that the values of the optimal solutions of the dual coincide. Then apply strong duality*)

b) Complementary Slackness and the Assignment Game

Problem 3: a) Consider the primal linear program

$$\begin{aligned} \max \quad & 3x_1 + 2x_2 + x_3 \\ \text{s.t.} \quad & x_1 + 2x_2 + x_3 \leq 4 \\ & 2x_1 + x_2 + 3x_3 \leq 6 \\ & x_1, x_2, x_3 \geq 0. \end{aligned}$$

For each of the following solution, use complementary slackness to decide whether they are optimal.

- $x^* = (\frac{8}{3}, \frac{2}{3}, 0)$;
- $\tilde{x} = (3, 0, 0)$.

b) Proceed as in part a) above, for the LP and its candidate solutions described below.

$$\begin{aligned} \text{maximize} \quad & z = 2x_1 + x_2 + x_3 \\ \text{subject to} \quad & x_1 + x_3 \leq 3 \\ & x_2 - x_3 \leq 2 \\ & x_1, x_2, x_3 \in \mathbb{R} \quad (\text{free}). \end{aligned}$$

- $x^* = (2, 3, 1)$.
- $\tilde{x} = (0, 0, 0)$.

Problem 4:

- a) Write with your favorite software a linear program that takes in input a number n of buyers, m of sellers, and their valuations, and outputs an equilibrium to the problem. Create 50 random instances with $n = 10$, $m = 20$ and compute the corresponding equilibria.
- b) Write code that decides, for an instance of the assignment game as in part a), if there is *another* equilibrium for the instance (i.e., one different from the one you have already found). Your code should solve one or more linear programs.
- c) Let (M^*, p^*) be an equilibrium to an instance I of the assignment game. Let $\overline{M}$ be a matching of maximum weight ($\overline{M} \neq M^*$) in the usual graph where nodes are agents, and the weight of an edge (i, j) (where i is a buyer and j is a seller) is the valuation of buyer i for the house of seller j . Is $(\overline{M}, p^*)$ always an equilibrium for I ?